1. **Research question and model purpose**

What are the suitable conditions for the wood ear mushroom (*Auricularia delicata*)? And does a South American monkey (*Callimico goeldii*) preferentially live in habitats with favorable conditions for wood ear mushroom growth?

Callimico goeldii are a small monkey native to the Western Amazon basin. They have been observed to spend up to 60% of their feeding time on mushrooms, especially the wood ear (*Auricularia delicata*). They also eat fruits, insects, flowers, and more but fungi takes up a large portion of their diet. Fungi are also patchy and rare, and seemingly don't have strong nutritional profiles. So why do *Callimico* consume so much? Are they preferentially choosing to live in habitats with abundant mushrooms, or are they simply exploiting this resource based on opportunity? Current hypotheses suggest that *Callimico* consume mushrooms as a way of avoiding interspecies competition with other monkeys, especially during the dry season when fruit is less abundant. This is supported by their large home ranges, much greater than other similar sized monkeys in the Amazon.

2. Occurrence data
Describe your proposed occurrence dataset:
• **Species or biological system**
Wood ear mushroom (*Auricularia delicata*)
Callitrichine monkey (*Callimico goeldii*)
• **Data source and citation or link**
Auricularia occurrences: https://www.gbif.org/occurrence/search?taxonKey=5VXWN
Callimico occurrences: https://www.gbif.org/occurrence/search?taxonKey=PQG3

• **Geographic and temporal coverage**
South America, Western Amazon Basin
I'm planning on using the limiting earliest record to 2026. I'm not sure if that's a good idea or not but it seems reasonable to only use records from when both species have been observed.

• **Approximate number of records available**
Auricularia has 1,990 occurrences
Callimico has 314 occurrences

• **Planned filters or quality-control decisions**
There are some preserved specimens for *Callimico* that I will filter out, they are in weird places like Texas, Maine, Germany. These monkeys would never actually live in those areas so all of the records should be from South America.
• **Important sampling biases or limitations**
Explain why the records are appropriate for your question and time period.
These records are applicable because they are the focal species of my question. I'm still

not too sure of how to define or choose the time period

3. Accessible area and background
Describe your current hypothesis about M, the area accessible to the species during the relevant
period. Explain:
• How you will define M geographically
• What biological, geographic, or historical reasoning supports that definition
• How background locations will be sampled within M
• One alternative definition of M that you could test

I know M is not the home range of a species but I think that is what I will use, or at least start with. The home range of *Callimico* is quite large for their size, so it must be a substantial portion of the actual M distance. However, I think they are mostly limited by the size of the forest fragments which they call home. For the mushroom, I'm really not sure because it is not as simple as a mammal. How does M work for a fungus? I would imagine they can't spread spores very far in the rainforest, and they rely on dead wood to propagate. I will have to do more research about that.

4. Environmental predictors
Identify approximately three to six candidate predictors. For each predictor, provide:
• Variable name
• Data source and version
• Units
• Spatial resolution
• Time period represented
• Biological reason for including it
• Whether it is a direct, indirect, or resource variable
Do not select variables because they are readily available. Explain how each predictor could
influence the organism or distribution.

1. Temperature
    a. Mushrooms in general are restricted to hotter climates. Mushroom cultivation usually involves warmer temperatures for optimal growth.
2. Precipitation/Humidity or maybe both?
    a. Mushroom growth is heavily dependent on water, but I'm not sure how to go about defining the predictor. Precipitation is obviously widely available but I feel that humidity or vapor pressure would be a better predictor. Would it hurt to have both?

3. Carbon biomass or CO_2
    a. Mushrooms need a source of organic carbon to survive. Wood ear mushrooms especially, grow on dead wood so they are very dependent on a carbon source.
4. Distance from water
    a. During the wet season, mushrooms probably receive enough rainfall to thrive but that changes during the dry season. There is less frequent rainfall so maybe proximity to bodies of water could influence mushroom abundance. But distance from water can't be an environmental layer so would I just download a raster of bodies of water?
5. Soil pH?
    a. This one might be a stretch but I read some studies about mushroom cultivation and pH seems to be a very important factor. However, wood ear mushrooms spend their entire life cycle on dead wood, so they don't need soil to survive. On the other hand, trees do need soil and the mushrooms are consuming the trees so maybe there is some connection.

5. Data compatibility and processing
Explain how you will evaluate and address:
• Coordinate reference systems
• Spatial extent and raster alignment
• Spatial resolution
• Temporal compatibility between occurrences and predictors
• Missing data
• Correlation among predictors
Identify any resampling, cropping, masking, or transformation you expect to perform.

I am skipping this section because I am not there yet in my planning and I want to discuss it in person.

6. Feasibility and uncertainty
Identify one data problem or modeling limitation you expect to run into and explain how you
might address it. Also, include any questions here on which you would like feedback.

I will probably be limited by the monkey occurrences. This is a very elusive species with observations in very select areas. How much do these occurrences actually capture the range and presence of this species? There are more occurrences than I expected but it's still not a lot so I hope it works out.